

BABU BANARASI DAS UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Student Management System Using MongoDB

Submitted to:

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Department: BCA (DS & AI)

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Submitted by:

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OBJECTIVES

- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

1. CREATE DATABASE

MongoDB Query:

```
use Students
```

Explanation: Creates or switches to the Student database.

```
test> use Students
switched to db Students
Students> █
```

2. CREATE COLLECTION

MongoDB Query:

```
db.createCollection("students")
```

Explanation: Creates a collection named students.

```
Students> db.createCollection("students")
{ ok: 1 }
Students> █
```

3. INSERT 50 STUDENTS

MongoDB Query:

```
db.bca23.insertMany([
  { student_id: 1, name: "Aryan Mehta", age: 19, gender: "Male",
    semester: 1, section: "A", email: "aryan.mehta@gmail.com",
    phone: "9123400001", city: "Lucknow", cgpa: 8.6 },
  { student_id: 2, name: "Diya Agarwal", age: 18, gender: "Female",
    semester: 1, section: "A", email: "diya.agarwal@gmail.com",
    phone: "9123400002", city: "Kanpur", cgpa: 9.0 },
  { student_id: 3, name: "Kabir Saxena", age: 19, gender: "Male",
    semester: 1, section: "B", email: "kabir.saxena@gmail.com",
    phone: "9123400003", city: "Lucknow", cgpa: 7.8 },
  { student_id: 4, name: "Ishita Srivastava", age: 19, gender: "Female",
    semester: 1, section: "B", email: "ishita.srivastava@gmail.com",
    phone: "9123400004", city: "Agra", cgpa: 8.0 },
  { student_id: 5, name: "Harshit Dubey", age: 20, gender: "Male",
    semester: 1, section: "A", email: "harshit.dubey@gmail.com",
```

```
phone: "9123400005", city: "Gorakhpur", cgpa: 7.1 },
{ student_id: 6, name: "Pari Chauhan", age: 18, gender: "Female",
  semester: 1, section: "A", email: "pari.chauhan@gmail.com",
  phone: "9123400006", city: "Lucknow", cgpa: 8.9 },
{ student_id: 7, name: "Lakshya Jaiswal", age: 19, gender: "Male",
  semester: 1, section: "B", email: "lakshya.jaiswal@gmail.com",
  phone: "9123400007", city: "Varanasi", cgpa: 7.4 },
{ student_id: 8, name: "Anika Rastogi", age: 19, gender: "Female",
  semester: 2, section: "A", email: "anika.rastogi@gmail.com",
  phone: "9123400008", city: "Bareilly", cgpa: 8.3 },
{ student_id: 9, name: "Shubham Tripathi", age: 20, gender: "Male",
  semester: 2, section: "B", email: "shubham.tripathi@gmail.com",
  phone: "9123400009", city: "Kanpur", cgpa: 7.6 },
{ student_id: 10, name: "Vaishnavi Pathak", age: 19, gender: "Female",
  semester: 2, section: "A", email: "vaishnavi.pathak@gmail.com",
  phone: "9123400010", city: "Lucknow", cgpa: 9.2 },
{ student_id: 11, name: "Tushar Bajpai", age: 20, gender: "Male",
  semester: 2, section: "B", email: "tushar.bajpai@gmail.com",
  phone: "9123400011", city: "Jhansi", cgpa: 8.0 },
{ student_id: 12, name: "Meera Kapoor", age: 19, gender: "Female",
  semester: 2, section: "A", email: "meera.kapoor@gmail.com",
  phone: "9123400012", city: "Delhi", cgpa: 8.7 },
{ student_id: 13, name: "Ayush Shukla", age: 21, gender: "Male",
  semester: 2, section: "B", email: "ayush.shukla@gmail.com",
  phone: "9123400013", city: "Lucknow", cgpa: 6.9 },
{ student_id: 14, name: "Bhavya Awasthi", age: 20, gender: "Female",
  semester: 3, section: "A", email: "bhavya.awasthi@gmail.com",
  phone: "9123400014", city: "Agra", cgpa: 8.5 },
{ student_id: 15, name: "Naman Chaturvedi", age: 20, gender: "Male",
  semester: 3, section: "B", email: "naman.chaturvedi@gmail.com",
  phone: "9123400015", city: "Meerut", cgpa: 7.9 },
{ student_id: 16, name: "Khushi Rawat", age: 19, gender: "Female",
  semester: 3, section: "A", email: "khushi.rawat@gmail.com",
  phone: "9123400016", city: "Kanpur", cgpa: 8.4 },
{ student_id: 17, name: "Parth Bhatia", age: 21, gender: "Male",
  semester: 3, section: "B", email: "parth.bhatia@gmail.com",
  phone: "9123400017", city: "Noida", cgpa: 7.3 },
{ student_id: 18, name: "Jiya Malhotra", age: 20, gender: "Female",
  semester: 3, section: "A", email: "jiya.malhotra@gmail.com",
  phone: "9123400018", city: "Lucknow", cgpa: 9.4 },
{ student_id: 19, name: "Sameer Ansari", age: 21, gender: "Male",
  semester: 3, section: "B", email: "sameer.ansari@gmail.com",
  phone: "9123400019", city: "Lucknow", cgpa: 8.1 },
{ student_id: 20, name: "Rhea Joshi", age: 20, gender: "Female",
  semester: 3, section: "A", email: "rhea.joshi@gmail.com",
  phone: "9123400020", city: "Varanasi", cgpa: 8.8 },
{ student_id: 21, name: "Gaurav Dixit", age: 21, gender: "Male",
  semester: 3, section: "B", email: "gaurav.dixit@gmail.com",
  phone: "9123400021", city: "Prayagraj", cgpa: 7.7 },
{ student_id: 22, name: "Anushka Tandon", age: 20, gender: "Female",
  semester: 4, section: "A", email: "anushka.tandon@gmail.com",
  phone: "9123400022", city: "Lucknow", cgpa: 8.6 },
{ student_id: 23, name: "Piyush Nigam", age: 21, gender: "Male",
  semester: 4, section: "B", email: "piyush.nigam@gmail.com",
  phone: "9123400023", city: "Kanpur", cgpa: 8.0 },
{ student_id: 24, name: "Sanya Saxena", age: 20, gender: "Female",
  semester: 4, section: "A", email: "sanya.saxena@gmail.com",
  phone: "9123400024", city: "Delhi", cgpa: 9.1 },
{ student_id: 25, name: "Ritesh Kushwaha", age: 21, gender: "Male",
  semester: 4, section: "B", email: "ritesh.kushwaha@gmail.com",
```

```
phone: "9123400025", city: "Ayodhya", cgpa: 7.0 },
{ student_id: 26, name: "Trisha Bansal", age: 20, gender: "Female",
  semester: 4, section: "A", email: "trisha.bansal@gmail.com",
  phone: "9123400026", city: "Agra", cgpa: 8.2 },
{ student_id: 27, name: "Lalit Sonkar", age: 22, gender: "Male",
  semester: 4, section: "B", email: "lalit.sonkar@gmail.com",
  phone: "9123400027", city: "Lucknow", cgpa: 7.5 },
{ student_id: 28, name: "Chhavi Mathur", age: 21, gender: "Female",
  semester: 4, section: "A", email: "chhavi.mathur@gmail.com",
  phone: "9123400028", city: "Noida", cgpa: 8.9 },
{ student_id: 29, name: "Mayank Rajput", age: 21, gender: "Male",
  semester: 4, section: "B", email: "mayank.rajput@gmail.com",
  phone: "9123400029", city: "Gorakhpur", cgpa: 8.3 },
{ student_id: 30, name: "Aarohi Trivedi", age: 20, gender: "Female",
  semester: 4, section: "A", email: "aarohi.trivedi@gmail.com",
  phone: "9123400030", city: "Kanpur", cgpa: 8.5 },
{ student_id: 31, name: "Utkarsh Pandey", age: 22, gender: "Male",
  semester: 5, section: "A", email: "utkarsh.pandey@gmail.com",
  phone: "9123400031", city: "Lucknow", cgpa: 7.9 },
{ student_id: 32, name: "Vidhi Bhardwaj", age: 21, gender: "Female",
  semester: 5, section: "B", email: "vidhi.bhardwaj@gmail.com",
  phone: "9123400032", city: "Prayagraj", cgpa: 9.3 },
{ student_id: 33, name: "Sarthak Goyal", age: 22, gender: "Male",
  semester: 5, section: "A", email: "sarthak.goyal@gmail.com",
  phone: "9123400033", city: "Meerut", cgpa: 7.2 },
{ student_id: 34, name: "Kritika Sinha", age: 21, gender: "Female",
  semester: 5, section: "B", email: "kritika.sinha@gmail.com",
  phone: "9123400034", city: "Lucknow", cgpa: 8.7 },
{ student_id: 35, name: "Anuj Katiyar", age: 22, gender: "Male",
  semester: 5, section: "A", email: "anuj.katiyar@gmail.com",
  phone: "9123400035", city: "Kanpur", cgpa: 8.0 },
{ student_id: 36, name: "Bhumika Arora", age: 21, gender: "Female",
  semester: 5, section: "B", email: "bhumika.arora@gmail.com",
  phone: "9123400036", city: "Delhi", cgpa: 8.4 },
{ student_id: 37, name: "Tejas Rathi", age: 21, gender: "Male",
  semester: 5, section: "B", email: "tejas.rathi@gmail.com",
  phone: "9123400037", city: "Varanasi", cgpa: 7.6 },
{ student_id: 38, name: "Manya Deshmukh", age: 21, gender: "Female",
  semester: 5, section: "A", email: "manya.deshmukh@gmail.com",
  phone: "9123400038", city: "Agra", cgpa: 9.0 },
{ student_id: 39, name: "Nitin Lodhi", age: 22, gender: "Male",
  semester: 5, section: "B", email: "nitin.lodhi@gmail.com",
  phone: "9123400039", city: "Ayodhya", cgpa: 6.8 },
{ student_id: 40, name: "Sonal Pal", age: 21, gender: "Female",
  semester: 5, section: "A", email: "sonal.pal@gmail.com",
  phone: "9123400040", city: "Lucknow", cgpa: 8.6 },
{ student_id: 41, name: "Ishaan Gaur", age: 22, gender: "Male",
  semester: 6, section: "A", email: "ishaan.gaur@gmail.com",
  phone: "9123400041", city: "Noida", cgpa: 8.2 },
{ student_id: 42, name: "Rashi Kesarwani", age: 21, gender: "Female",
  semester: 6, section: "B", email: "rashi.kesarwani@gmail.com",
  phone: "9123400042", city: "Kanpur", cgpa: 8.9 },
{ student_id: 43, name: "Prateek Vishwakarma", age: 22, gender: "Male",
  semester: 6, section: "A", email: "prateek.vishwakarma@gmail.com",
  phone: "9123400043", city: "Lucknow", cgpa: 7.8 },
{ student_id: 44, name: "Zoya Siddiqui", age: 21, gender: "Female",
  semester: 6, section: "B", email: "zoya.siddiqui@gmail.com",
  phone: "9123400044", city: "Lucknow", cgpa: 9.1 },
{ student_id: 45, name: "Ankit Maurya", age: 22, gender: "Male",
  semester: 6, section: "A", email: "ankit.maurya@gmail.com",
```

```

    phone: "9123400045", city: "Prayagraj", cgpa: 7.4 },
  { student_id: 46, name: "Charvi Gill", age: 21, gender: "Female",
    semester: 6, section: "B", email: "charvi.gill@gmail.com",
    phone: "9123400046", city: "Delhi", cgpa: 8.3 },
  { student_id: 47, name: "Rohit Baghel", age: 22, gender: "Male",
    semester: 6, section: "A", email: "rohit.baghel@gmail.com",
    phone: "9123400047", city: "Agra", cgpa: 8.0 },
  { student_id: 48, name: "Naina Kohli", age: 21, gender: "Female",
    semester: 6, section: "B", email: "naina.kohli@gmail.com",
    phone: "9123400048", city: "Gorakhpur", cgpa: 8.8 },
  { student_id: 49, name: "Devansh Rai", age: 22, gender: "Male",
    semester: 6, section: "A", email: "devansh.rai@gmail.com",
    phone: "9123400049", city: "Kanpur", cgpa: 7.7 },
  { student_id: 50, name: "Tanisha Chopra", age: 21, gender: "Female",
    semester: 6, section: "B", email: "tanisha.chopra@gmail.com",
    phone: "9123400050", city: "Lucknow", cgpa: 9.2 }
])

```

Explanation: Inserts multiple student documents into the collection.

```

Students> db.bca23.insertMany([ ...50 student documents... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66f8000052e6b438f2a7'),
    '1': ObjectId('66f800001269e0d376513'),
    '2': ObjectId('66f80002a6a3a4500c5c'),
    ...
  }
}
[ Output shortened for the report: insertedIds '0' to '49' exist; first three shown ]
Students> █

```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.bca23.find()
```

Explanation: Retrieves documents from the students collection.

```

Students> db.bca23.find()
[
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cgpa: 9
  },
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  ... 47 more items
]
[ Output shortened for the report: showing 3 of 50 documents ]
Students> █

```

5. EQUAL TO — \$eq

MongoDB Query:

```
db.bca23.find({cgpa: {$eq: 8}})
```

Explanation: Finds students whose cgpa are exactly 8.

```

Students> db.bca23.find({cgpa: {$eq: 8}})
[
  {
    _id: ObjectId('66f80003128b2f33d23f'),
    student_id: 4,
    name: 'Ishita Srivastava',
    age: 19,
    gender: 'Female',
    semester: 1,
    section: 'B',
    email: 'ishita.srivastava@gmail.com',
    phone: '9123400004',
    city: 'Agra',
    cgpa: 8
  },
  {
    _id: ObjectId('66f8000a11e20b8f3d9c'),
    student_id: 11,
    name: 'Tushar Bajpai',
    age: 20,
    gender: 'Male',
    semester: 2,
    section: 'B',
    email: 'tushar.bajpai@gmail.com',
    phone: '9123400011',
    city: 'Jhansi',
    cgpa: 8
  },
  {
    _id: ObjectId('66f80016dbc496cb2217'),
    student_id: 23,
    name: 'Piyush Nigam',
    age: 21,
    gender: 'Male',
    semester: 4,
    section: 'B',
    email: 'piyush.nigam@gmail.com',
    phone: '9123400023',
    city: 'Kanpur',
    cgpa: 8
  },
  ... 2 more items
]
[ Output shortened for the report: showing 3 of 5 documents ]
Students> █

```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.bca23.find({cgpa: {$gt: 8.5}})
```

Explanation: Finds students whose cgpa are greater than 8.5.

```

Students> db.bca23.find({cpga: {$gt: 8.5}})
[
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cpga: 8.6
  },
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cpga: 9
  },
  {
    _id: ObjectId('66f800055d9dc9f89531'),
    student_id: 6,
    name: 'Pari Chauhan',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'pari.chauhan@gmail.com',
    phone: '9123400006',
    city: 'Lucknow',
    cpga: 8.9
  },
  ... 15 more items
]
[ Output shortened for the report: showing 3 of 18 documents ]
Students> █

```

7. LESS THAN — \$lt

MongoDB Query:

```
db.bca23.find({age: {$lt: 20}})
```

Explanation: Finds students whose age are less than 20.

```

Students> db.bca23.find({age: {$lt: 20}})
[
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cgpa: 9
  },
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  ... 7 more items
]
[ Output shortened for the report: showing 3 of 10 documents ]
Students> █

```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.bca23.find({age: {$gte: 20}})
```

Explanation: Finds students whose age are 20 or above.

```

Students> db.bca23.find({age: {$gte: 20}})
[
  {
    _id: ObjectId('66f80004892f902b1818'),
    student_id: 5,
    name: 'Harshit Dubey',
    age: 20,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'harshit.dubey@gmail.com',
    phone: '9123400005',
    city: 'Gorakhpur',
    cgpa: 7.1
  },
  {
    _id: ObjectId('66f80008099950d81600'),
    student_id: 9,
    name: 'Shubham Tripathi',
    age: 20,
    gender: 'Male',
    semester: 2,
    section: 'B',
    email: 'shubham.tripathi@gmail.com',
    phone: '9123400009',
    city: 'Kanpur',
    cgpa: 7.6
  },
  {
    _id: ObjectId('66f8000a11e20b8f3d9c'),
    student_id: 11,
    name: 'Tushar Bajpai',
    age: 20,
    gender: 'Male',
    semester: 2,
    section: 'B',
    email: 'tushar.bajpai@gmail.com',
    phone: '9123400011',
    city: 'Jhansi',
    cgpa: 8
  },
  ... 37 more items
]
[ Output shortened for the report: showing 3 of 40 documents ]
Students> █

```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.bca23.find({cgpa: {$lte: 8.5}})
```

Explanation: Finds students whose cgpa are 8.5 or below.

```

Students> db.bca23.find({cgpa: {$lte: 8.5}})
[
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  {
    _id: ObjectId('66f80003128b2f33d23f'),
    student_id: 4,
    name: 'Ishita Srivastava',
    age: 19,
    gender: 'Female',
    semester: 1,
    section: 'B',
    email: 'ishita.srivastava@gmail.com',
    phone: '9123400004',
    city: 'Agra',
    cgpa: 8
  },
  {
    _id: ObjectId('66f80004892f902b1818'),
    student_id: 5,
    name: 'Harshit Dubey',
    age: 20,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'harshit.dubey@gmail.com',
    phone: '9123400005',
    city: 'Gorakhpur',
    cgpa: 7.1
  },
  ... 29 more items
]
[ Output shortened for the report: showing 3 of 32 documents ]
Students> █

```

10. UPDATE OPERATION

MongoDB Query:

```

db.bca23.updateOne(
  {student_id:5},
  {$set:{cgpa:9.5}}
)

```

Explanation: Updates the cgpa to 9.5 of the student with student_id: 5.

```

Students> db.bca23.updateOne(
  ...   {student_id:5},
  ...   {$set:{cgpa:9.5}}
  ... )
{
  acknowledged: true,
  insertedId: None,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Students> █

```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.bca23.deleteOne({student_id:10})
```

Explanation: Deletes one document matching student_id: 10.

```
Students> db.bca23.deleteOne({student_id:10})
{
  acknowledged: true,
  deletedCount: 1
}
Students> █
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.bca23.deleteMany({
  cgpa: {$lt:7.5}
})
```

Explanation: Deletes all students having cgpa below 7.5.

```
Students> db.bca23.deleteMany({
...   cgpa: {$lt:7.5}
... })
{
  acknowledged: true,
  deletedCount: 7
}
Students> █
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.bca23.find({
  $and:[
    {cgpa:{$gt:8}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions (cgpa above 8 and age 19).

```

Students> db.bca23.find({
...   $and:[
...     {cgpa:{$gt:8}},
...     {age:19}
...   ]
... })
[
  {
    _id: ObjectId('66f800052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f8000781e74ef536f6'),
    student_id: 8,
    name: 'Anika Rastogi',
    age: 19,
    gender: 'Female',
    semester: 2,
    section: 'A',
    email: 'anika.rastogi@gmail.com',
    phone: '9123400008',
    city: 'Bareilly',
    cgpa: 8.3
  },
  {
    _id: ObjectId('66f8000b1738f7d98d11'),
    student_id: 12,
    name: 'Meera Kapoor',
    age: 19,
    gender: 'Female',
    semester: 2,
    section: 'A',
    email: 'meera.kapoor@gmail.com',
    phone: '9123400012',
    city: 'Delhi',
    cgpa: 8.7
  },
  ... 1 more items
]
[ Output shortened for the report: showing 3 of 4 documents ]
Students> █

```

14. OR OPERATION — \$or

MongoDB Query:

```

db.bca23.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})

```

Explanation: Finds students from Lucknow or Kanpur.

```

Students> db.bca23.find({
...   $or:[
...     {city:"Lucknow"},
...     {city:"Kanpur"}
...   ]
... })
[
  {
    _id: ObjectId('66f800052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cgpa: 9
  },
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  ... 18 more items
]
[ Output shortened for the report: showing 3 of 21 documents ]
Students> █

```

15. SORT OPERATION

A. MongoDB Query:

```
db.bca23.find().sort({age : 1})
```

Explanation: Sorts students by age. Use 1 for ascending order.

```

Students> db.bca23.find().sort({age : 1})
[
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cgpa: 9
  },
  {
    _id: ObjectId('66f800055d9dc9f89531'),
    student_id: 6,
    name: 'Pari Chauhan',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'pari.chauhan@gmail.com',
    phone: '9123400006',
    city: 'Lucknow',
    cgpa: 8.9
  },
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  ... 39 more items
]
[ Output shortened for the report: showing 3 of 42 documents ]
Students> █

```

B. MongoDB Query:

```
db.bca23.find().sort({semester: -1})
```

Explanation: Sorts students by semester. Use -1 for descending order.

```

Students> db.bca23.find().sort({semester: -1})
[
  {
    _id: ObjectId('66f800287403e4305c90'),
    student_id: 41,
    name: 'Ishaan Gaur',
    age: 22,
    gender: 'Male',
    semester: 6,
    section: 'A',
    email: 'ishaan.gaur@gmail.com',
    phone: '9123400041',
    city: 'Noida',
    cgpa: 8.2
  },
  {
    _id: ObjectId('66f800294cbd87ad3f98'),
    student_id: 42,
    name: 'Rashi Kesarwani',
    age: 21,
    gender: 'Female',
    semester: 6,
    section: 'B',
    email: 'rashi.kesarwani@gmail.com',
    phone: '9123400042',
    city: 'Kanpur',
    cgpa: 8.9
  },
  {
    _id: ObjectId('66f8002acb5c74272e05'),
    student_id: 43,
    name: 'Prateek Vishwakarma',
    age: 22,
    gender: 'Male',
    semester: 6,
    section: 'A',
    email: 'prateek.vishwakarma@gmail.com',
    phone: '9123400043',
    city: 'Lucknow',
    cgpa: 7.8
  },
  ... 39 more items
]
[ Output shortened for the report: showing 3 of 42 documents ]
Students> █

```

16. LIMIT OPERATION

MongoDB Query:

```
db.bca23.find().limit(5)
```

Explanation: Displays only the first five documents.

```

Students> db.bca23.find().limit(5)
[
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f80001269e0d376513'),
    student_id: 2,
    name: 'Diya Agarwal',
    age: 18,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'diya.agarwal@gmail.com',
    phone: '9123400002',
    city: 'Kanpur',
    cgpa: 9
  },
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  ... 2 more items
]
[ Output shortened for the report: showing 3 of 5 documents ]
Students> █

```

17. SEARCH BY GENDER

MongoDB Query:

```
db.bca23.find({gender:"Male"})
```

Explanation: Finds all students whose gender is male.

```

Students> db.bca23.find({gender:"Male"})
[
  {
    _id: ObjectId('66f8000052e6b438f2a7'),
    student_id: 1,
    name: 'Aryan Mehta',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aryan.mehta@gmail.com',
    phone: '9123400001',
    city: 'Lucknow',
    cgpa: 8.6
  },
  {
    _id: ObjectId('66f80002a6a3a4500c5c'),
    student_id: 3,
    name: 'Kabir Saxena',
    age: 19,
    gender: 'Male',
    semester: 1,
    section: 'B',
    email: 'kabir.saxena@gmail.com',
    phone: '9123400003',
    city: 'Lucknow',
    cgpa: 7.8
  },
  {
    _id: ObjectId('66f80004892f902b1818'),
    student_id: 5,
    name: 'Harshit Dubey',
    age: 20,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'harshit.dubey@gmail.com',
    phone: '9123400005',
    city: 'Gorakhpur',
    cgpa: 9.5
  },
  ... 15 more items
]
[ Output shortened for the report: showing 3 of 18 documents ]
Students> █

```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.bca23.countDocuments()
```

Explanation: Counts documents currently present in the collection (performing after the delete function).

```

Students> db.bca23.countDocuments()
42
Students> █

```

MONGODB OPERATORS USED:

Operator	Meaning	Example
\$eq	Equal to	cgpa: {\$eq:8}
\$gt	Greater than	cgpa: {\$gt:8.5}
\$lt	Less than	age: {\$lt:20}
\$gte	Greater than or equal to	age: {\$gte:20}
\$lte	Less than or equal to	cgpa: {\$lte:8.5}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.